

CredVeritas: Jurisdiction-Aware Credit Compliance Layer

Safeguarding DBS Bank's Algorithmic Parity Across
the US-Singapore Lending Corridor

- Prepared For: Senior Management & Chief Compliance Officer (CCO), DBS Bank Ltd.
- Strategic Objective: Seamlessly decoupling cross-border regulatory compliance from core underwriting logic.
- Core Approach: Deploying an independent real-time audit layer utilizing Explainable AI (XAI) and "Shadow Labeling."

The Compliance Split: Individual Justice vs. Systemic Fairness

US Jurisdiction (CFPB Regulation B / ECOA):

- Focus: Micro-level individual procedural justice.
- Mandate: Every automated credit denial requires an immediate, human-readable Adverse Action Notice isolating the top 3-4 quantitative reasons. Black-box institutional defenses are prohibited.

Singapore Jurisdiction (MAS FEAT Principles & AI MRM):

- Focus: Macro-level portfolio-wide systemic fairness.
- Mandate: Continuous statistical group parity auditing to eliminate systemic proxy discrimination and demographic "de-banking."

The Technical Solution ("How"):

CredVeritas leaves the core underwriting model untouched but uses "Shadow Labels" in the compliance layer to dynamically switch rules based on geographic origin (Geo-IP).

Moving Beyond Static Checklists: From Reactive to Real-Time Audit

The Failure of Spreadsheets & Memos:

Traditional compliance relies on retrospective annual sampling or static Excel checklists. In a high-frequency digital banking environment, this is equivalent to "protecting the crime scene months after the incident."

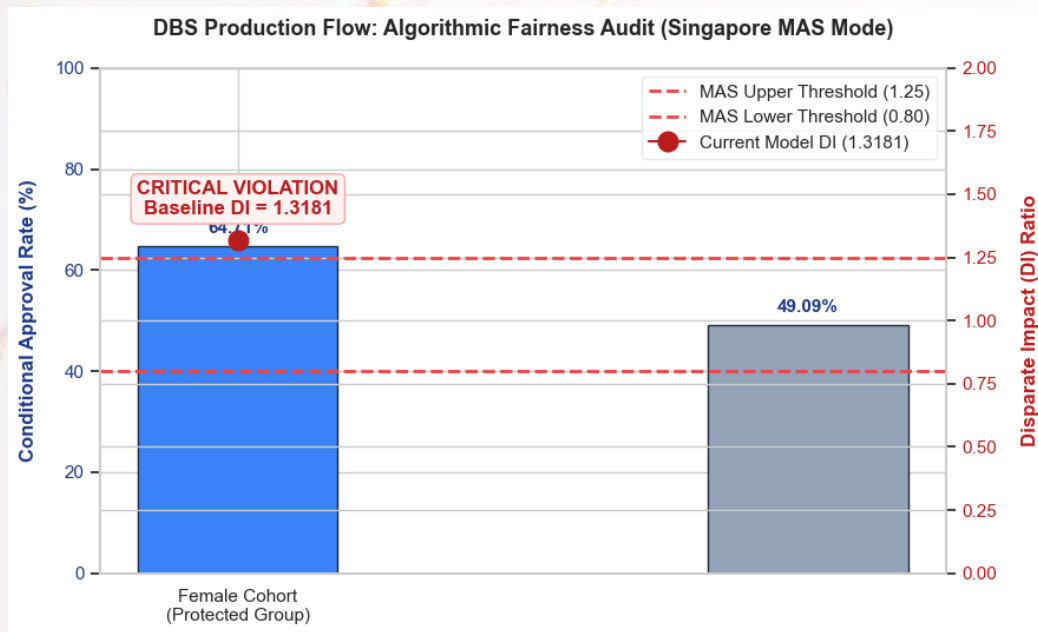
Real-Time Transactional Stream Scanning:

CredVeritas acts as an inline automated compliance filter, parsing production transactions the millisecond they occur.

Strategic Alignment for DBS:

Maximizes DBS' s digital-first banking efficiency while eliminating the multi-million dollar liability risks associated with cross-border algorithmic non-compliance.

Empirical Proof: Detecting Systemic Bias in Live Streams



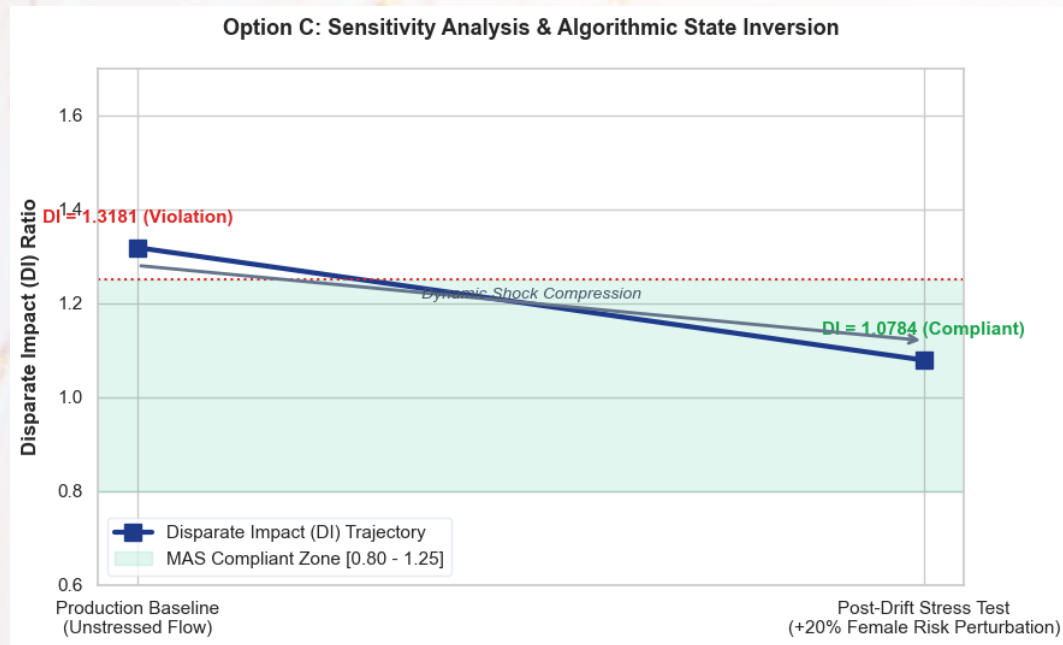
Live Stream Audit Results :

- Captured 34 US loan denials and dynamically generated localized Adverse Action letters (e.g., isolating credit-to-duration risk for ID 240 and asset insufficiency for ID 919).

Singapore Portfolio Metric Breach:

- Instantly flagged a Disparate Impact (DI) Ratio of 1.3181 in Singapore Mode, crossing the maximum regulatory boundary of 1.25.
- Visual Evidence: As shown in the chart, the baseline model executes a silent proxy discrimination, triggering a Critical Compliance Violation for over-rejecting the male cohort.

Option C Sensitivity Analysis: The Danger of Static Compliance



Macroeconomic Distortion Simulation:

Injected a +20% risk shock on the female demographic cohort to simulate macro-drift.

The Parity Volatility Inversion:

The DI Ratio counter-intuitively compressed from a non-compliant 1.3181 down to a compliant 1.0784 (in $[0.80, 1.25]$).

Crucial Risk Management Insight:

A static year-end report would celebrate 1.0784 as perfectly compliant, completely missing the fact that the model has severely drifted under economic stress.

Defining the Compliance Frontier: Why Core Judgment Cannot Be Automated

What the Tool Does NOT Do:

CredVeritas strictly does NOT automatically alter credit model weights, modify risk parameters, or autonomously override loan approval decisions.

The Rationale for the Boundary (The Risk Weighting):

Compliance does not equal mechanical parity. Forcing an AI to automatically manipulate its scoring criteria to satisfy a mathematical ratio would distort credit risk boundaries and directly compromise DBS's asset safety.

Human-in-the-Loop Sovereignty:

The tool diagnostics and issues alerts; however, core lending Judgment cannot be outsourced. Strategic model recalibration remains the sovereign commercial right of the DBS Credit Committee.

PoC Limitation:

We honestly acknowledge that the 72-record PoC carries statistical noise; scaled integration is required to maximize DI calculation precision.



THANKS!